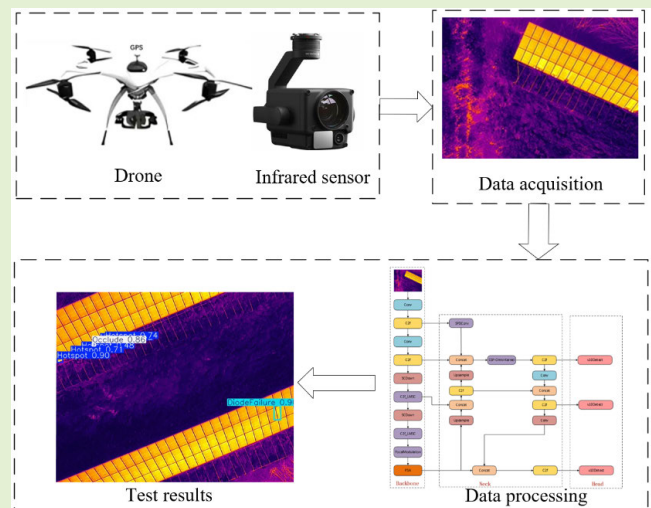


LFS-YOLO: A PV Panel Defect Detection Algorithm for Drone Infrared Sensors

Ziqi Ma^{ID}, Hanrui Guo, Hao Wen, and Yingli Cao^{ID}

Abstract—Photovoltaic (PV) panel hot spot defects will reduce the power generation capacity of PV panels, which will seriously threaten the safe operation of the power station, and hot spot defect detection and timely repair is an important work of PV power plant operation and maintenance. As a key technology to improve the autonomous perception ability of drone infrared sensors, object detection has become the focus of drone inspection in PV power plants. In this article, a hot spot defect detection algorithm according to infrared images of aerial PV is proposed for practical engineering problems such as defects with different morphology, unclear boundary features, and small target defects. First, a multi-scale lightweight convolution module C2f_LMSC is designed to realize multiscale feature extraction and reduce the number of parameters and computation amount of the model. Second, the Focal Nets are used to replace the original spatial pyramid pooling-fast (SPPF) module, which improves the model's ability to capture the features of the defects with inconspicuous boundary features. Furthermore, in order to improve the model's effectiveness in detecting small target defects, a feature pyramid small object resultful pyramid (SORP) is designed for small target detection. In addition, in order to improve the detection effect of the model on small target defects, a feature pyramid SORP for small target detection is designed. The test results show that the average accuracy of LFS-YOLO model detection is improved by 3.2%–91.9% compared with the pre-improvement period, the defect leakage phenomenon existed in the original model is improved, and the research results can provide technical support for the detection of PV defects by drones.

Index Terms—Defect detection, hot spot, infrared sensors, photovoltaic (PV) panel, YOLOv10.



I. INTRODUCTION

WITH the world's energy needs continuing to rise and environmental preservation becoming more widely recognized, China's energy supply system is transitioning from traditional fossil fuels to renewable and green energy [1], [2], [3]. At the United Nations Climate Change Conference in Madrid in 2019, China released China's photovoltaic (PV) Development Outlook 2050, which proposed that China's installed PV capacity will reach 5000 GW in 2050 and that the average annual proportion of solar power generation will exceed 40% of the total power generation, becoming China's

number one source of electricity [4]. In this context, China's PV power generation industry is developing rapidly, and the PV erection area is growing rapidly, along with the increasingly severe maintenance pressure of PV power plant [5], [6], [7]. PV runs in the outdoor environment for a long time, prone to various defects, and the defective parts of the PV in the operation process will become the load in the circuit, consume the energy generated by other parts of the normal work, and consume it in the form of heat, resulting in the defective parts of the local temperature rises and producing the hot spot effect [8], [9], [10]. This not only makes the PV panel's power generation capacity greatly reduced but, if not timely maintenance, will also lead to the whole PV panel damage and even cause a fire [11], [12], [13]. Therefore, defect detection of PV has important research value and significance for the PV power generation industry.

China's PV power station for PV hot spot defect detection is mainly divided into manual inspection and drone inspection two ways [14]. As PV power stations are generally larger and the environment is caused by uncertainties, the use of manual inspection is time-consuming and laborious, not only

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low inspection efficiency but also may jeopardize the personal safety of workers [15]. The drone has a low cost, wide field of view, is not easily constrained by the terrain of the terrain, and can be easily adapted to work on rooftops, hillsides, lakes, and other environments. In recent years, the application of this technology has been gradually extended to the operation and maintenance inspection of PV power plants. [16].

In the context of PV panel defect detection, the utilization of an infrared sensor facilitates the acquisition of an infrared image of the PV panel during its operational process, thereby enabling the visualization of temperature distribution. The presence of abnormal heating results in a significant enhancement of the brightness of the infrared image on the defective parts compared to the normal parts. This enhancement renders traditional visible light detection ineffective in identifying defects within PV cells, such as diode short circuits and other hidden defects [17], [18]. The advent of computer vision technology has led to the widespread utilization of target detection algorithms based on deep learning in various domains [19], [20]. The integration of the drone with the PV panel to acquire infrared image data is a recent development in the field. The incorporation of these data into a target detection model enables the utilization of a convolutional neural network (CNN) architecture to detect thermal anomalies in the infrared image. This approach facilitates intelligent classification and localization, as well as the automatic identification of various defect types, thereby establishing a defect characteristics database [21], [22], [23]. The noncontact and high-sensitivity characteristics of the system facilitate automation and high precision in the detection of defects in PV panels. This capability addresses the limitations of traditional manual inspection methods, which often result in low efficiency and high defect leakage rates. Additionally, the system provides data-driven decision-making support for the operation and maintenance of PV power stations, enhancing the reliability of PV power generation systems [24], [25].

Presently, artificial neural networks are extensively employed for the purpose of defect detection in PV panels, a practice that is predicated on the proven efficacy of these networks in the domains of image processing and target detection. Zhao et al. [26] designed a fine-graph inference network consisting of a backbone network, cross-embedded fine-graph inference modules, and a prediction head, which improves the defect detection accuracy in complex scenarios of PV power plants by constructing structured relational models of multihigh-height hot spot defects. However, the ability of the RGR-Net to recognize weak defects under extreme lighting conditions is yet to be verified. Yin et al. [27] developed an infrared image defect detection algorithm for PV panels based on an enhanced version of the YOLOv7 model. This modification led to an improvement in the detection accuracy of the original model. However, the model size proved to be a significant challenge, necessitating further investigation to optimize its performance and scalability. Lei et al. [28] proposed a combination of segmentation and detection of infrared image PV panel hot spot defect detection method for PV power plants. This approach is

particularly relevant in the context of the complex background of the operation and maintenance of PV power plants, where difficulties in detecting hot spots in PV panels can be effectively addressed. However, it is important to note that this method has certain limitations in terms of detecting small targets. Hong et al. [29] proposed an infrared image detection algorithm for PV power plants based on CEMP-YOLOv10n. They designed the ABCG_Block structure and optimized the C2f structure within the backbone component to enhance the feature extraction capability. However, they did not take into account the blurring characteristics of defect edges. Jia et al. [30] proposed an enhanced varifocal net method to detect defects in electro luminescence (EL) images of PV panels. However, the classification ability is not strong due to the different distribution shapes and locations of defective targets in PV modules. Barrett et al. [31] designed a YOLOv8s-seg model based on YOLOv8. The model achieved an average accuracy of 87% in the segmentation of solar panels and did not detect defects on PV panels. In their study, Segovia Ramírez et al. [32] propose a methodology employing two successive CNNs to detect hot spots in IoT platforms and reduce the number of false alarms. However, the applicability of this methodology is limited to a wide range of temperatures, and it lacks the capacity to recognize defects with insignificant differences. Tang et al. [33] put forward a CNN model depth reduction method based on migration learning. This method was validated with the VGG16 model, and a multiscale feature extraction module was developed to reduce the model's complexity and improve its classification performance. However, the model's overall accuracy was only improved by 0.98%. Cui et al. [34] employ a dust feature enhancement module based on the hue, lightness, saturation (HLS) color space in conjunction with Mask region-based CNN (R-CNN) to detect dusty regions. In order to target the unique environment of offshore floating solar power plants, this method reduces the misclassification and omission of dusty regions. However, the targeting environment is too homogeneous, and the model lacks generalization.

The above methods are mostly based on the convolution operation of fixed sensory fields, which limits the ability of multiscale feature capture under the constraint of low computational volume, and it is difficult to adapt to the characteristics of the PV panel defects with different morphologies; the fixed pooling kernel of the traditional spatial pyramid pooling-fast (SPPF) module has low sensitivity to the boundary information of the gradual thermal radiation region, and it is difficult to identify the boundary features of the defects that are not obvious. Although directly adding a small target detection layer or attention module can enhance the model's response to a specific defective region, it is difficult to distinguish multiple neighboring small targets in this way, which leads to misdetection and omission of small target defects and also results in excessive model computation and more time-consuming postprocessing, which is not conducive to the deployment in mobile devices.

In this study, an improved LFS-YOLO model for detecting common hot spot defects in PV is proposed based on the task requirements for real-time inspection of drone infrared

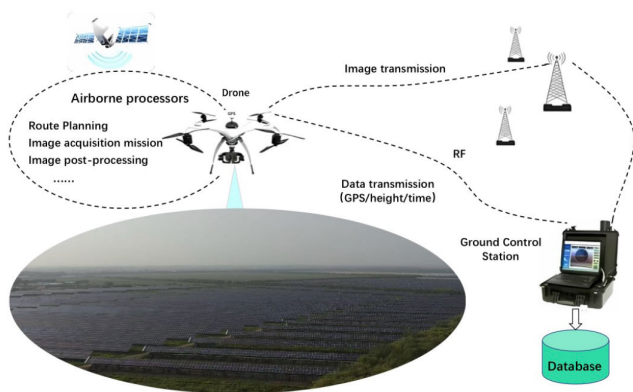


Fig. 1. Data acquisition.

sensors. First, we propose to design a multiscale lightweight convolution module that operates at multiple scales to substitute certain components of the C2f module within the backbone network, enabling the model to extract features at various scales. Second, we improve the SPPF module to improve the model's ability to recognize the defects of the boundary features that are not obvious. Finally, we design the feature pyramid for the improvement of the neck network to tackle the issue of easily losing small target features and the associated detection omissions.

II. DATA ACQUISITION AND PROCESSING

A. Image Data Acquisition

The infrared image data of the PV used in this study were collected from a PV power station in the northern area of Shenyang City, Liaoning Province, which has a total installed capacity of 20 MW and a total land area of 1100 acres. Using DJI Jingwei M350RTK drone equipped with Zenith H20T sensor to collect infrared images of PV at a height of 50 m, with an image resolution of 640×512 , taken on August 3, 2024, a total of 2188 PV infrared images were collected, and a total of 1568 infrared images containing defects in PV were screened out for network training and testing. The data acquisition is shown in Fig. 1.

B. Defect Detection Tasks and Characterization

During the data collection process, it was found that there are three main types of hot spot defects in this PV plant, as shown in Fig. 2, which are solar cell defects, diode failure, and occlude. Solar cell defects as shown in Fig. 2(a), in the infrared image for the shape and size of the dot-type hot spot, caused by solar cell defects are mainly due to the PV panel cracks, cracks, internal damage, and so on, since the existence of such defects is too small, in the detection process that is easy to cause the phenomenon of leakage detection. The diode defect, shown in Fig. 2(b), manifests itself as a regular rectangular hot spot on the PV panel, and the cause of the defect is the breakdown of the diode in the junction box on the back of the module. As shown in Fig. 2(c), manifested as dark irregularly shaped foreign objects on the PV panels, this situation is mostly caused by the masking of bushes, trees, and so on, which should not only be considered as

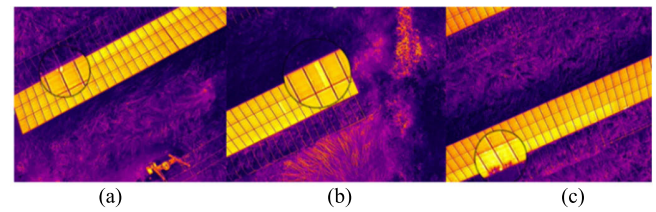


Fig. 2. Defect type. (a) Solar cell defects. (b) Diode failure. (c) Occlude.

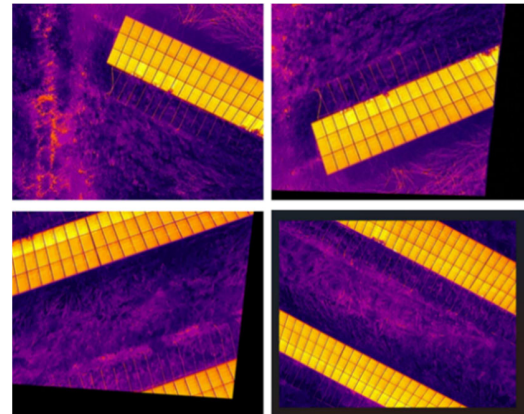


Fig. 3. Part of the data-enhanced images.

a maintenance treatment of the PV panels themselves but also remove the masking objects in time to avoid causing subsequent problems. Due to the irregular shape of the occlude and the varying shading effect of the edges of the scrub trees, this kind of defect presents unclear edge features, which may easily result in the difficulty of the model to accurately extract the global features during the detection procedure. As observed in the visualization image, the morphology of various types of defects is different, and the size is very small; some small targets are easy to lose their features during the model learning process, and the recognition difficulty is high. In this article, the proposed detection method will be utilized to detect hot spot defects on datasets with such characteristics.

C. Image Annotation and Dataset Construction

The solar cell defects, diode failure, and occlude are manually labeled using the image labeling tool Labeling (v 1.8.6) to obtain a txt label document containing data on the annotated frames, which is distributed with a 7:2:1 proportion for the training set, test set, and validation set. To mitigate the risk of overfitting that arises from having a small sample size and to bolster the generalization abilities of the model, we implement data enhancement methods such as random translations, rotations, flipping, and compression on the samples and labels, ensuring that the proportions of the training, validation, and test datasets stay the same. Some of the pictures are presented in Fig. 3. An overall count of 4704 images and their corresponding labeled files in the txt format are obtained after expansion. Among them, 3293 images are in the training set, 941 images are in the validation set, and 470 images are in the test set.

TABLE I

COMPARISON TEST RESULTS OF DIFFERENT MODEL PERFORMANCE

Model	mAP50%	R%	Params/M	Size/MB
Faster-RCNN	84.2	82.5	13.7	25.2
SSD	82.3	81.7	7.2	14.6
YOLOv8n	85.9	84.8	3.0	6.3
YOLOv9t	83.5	83.4	2.8	4.7
YOLOv10n	88.7	86.0	2.3	5.2

III. YOLOV10N MODEL AND PRETESTING

The YOLO series of object detection models has drawn considerable attention because of their remarkable real-time performance and high accuracy in detection. This lineup of models creatively utilizes a single-stage framework, eliminating the complicated step of generating region candidate boxes, which greatly decreases computational demands and allows for implementation on embedded devices with restricted computational capabilities. YOLOv10 is an optimized and improved model based on YOLOv8. Compared to YOLOv9 and YOLOv8, YOLOv10 eliminates the dependence on NMS by introducing a consistent dual assignment strategy that combines both one-to-many and one-to-one prediction strategies during training. This improvement significantly reduces the computational overhead in the inference process and improves the efficiency of the model's operation on hardware, striking a good balance between detection speed and accuracy to detect targets quickly and accurately, making it more suitable for deployment on mobile devices. In addition, YOLOv10 has a strong generalization ability to better adapt to different inspection tasks and different scenarios and presently comprises the n, s, m, l, and x versions, with the size and the number of parameters increasing in turn. Since this study is oriented to drone infrared sensor-based PV panel defect detection applications, we choose to use the YOLOv10n model with the smallest size and conduct comparison tests with other mainstream target detection models, and the test results are shown in Table I.

YOLOv10 is currently available in several versions: n, s, m, l, and x, with each version featuring progressively larger dimensions and parameters. The architecture is structured into three fundamental parts: the backbone network (Backbone), the neck network (Neck), and the detection head (Head). As illustrated in Fig. 4, the architecture of the YOLOv10n network is described. The Backbone is tasked with feature identification and is comprised of three primary modules: Conv, C2f, and SPPF. The Neck integrates feature representations of varying measurements obtained from the Backbone, allowing for the fusion of surface-level and deep features. Meanwhile, the Head employs a decoupled approach to handle classification, regression, and confidence tasks independently before merging the results.

The outcomes of detection for the YOLOv10n model on the constructed dataset are shown in Fig. 5. From the figure, it can be seen that the defects have missed detection due to their different morphology and size, unclear defect edges, and more tiny defects; the detection confidence is generally not high and there are defects in the missed detection phenomenon, so it is

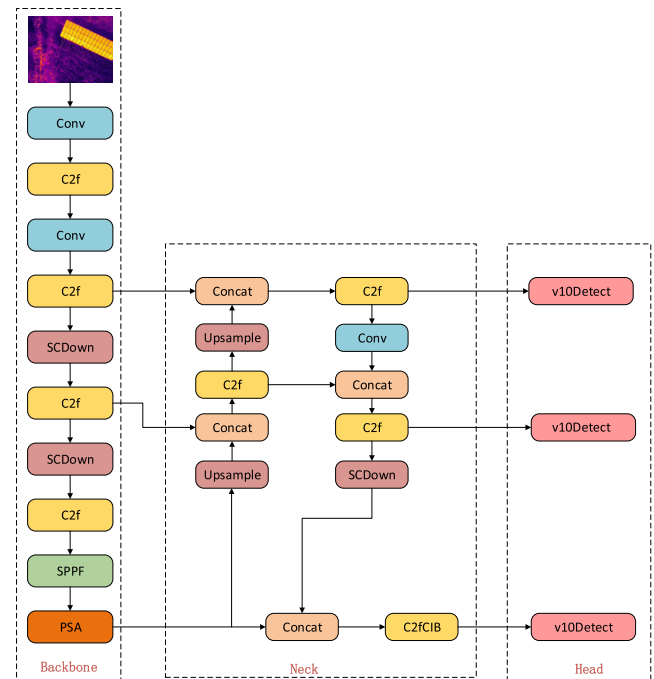


Fig. 4. Before improvement—YOLOv10n.

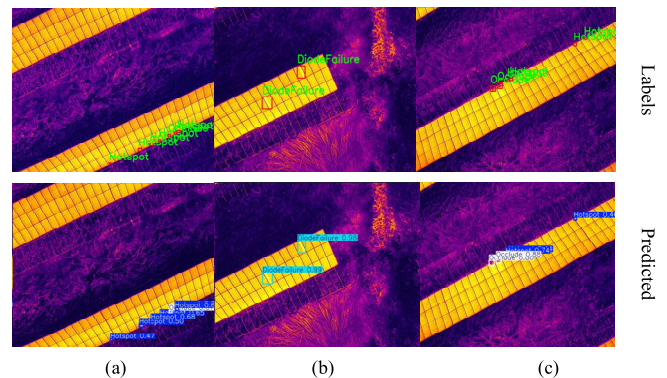


Fig. 5. Detection results of YOLOv10n on different defects. (a) PV cell. (b) Diode failure. (c) Occlude.

necessary to improve the network model for such problems to enhance the detection effect.

IV. CONSTRUCTION OF LFS-YOLO MODEL

Through the pre-experiment, it can be found that the YOLOv10n model has a poor detection effect due to defects with different morphologies, unclear edges, and a large number of tiny defects. Therefore, three improvements are made as follows: 1) a multiscale lightweight convolution module C2f_LMSC (light multiscale conv) is designed to replace part of the C2f module in the backbone network to realize multiscale feature extraction and reduce the number of model parameters and computation; 2) replace the original SPPF module with focal modulation nets, which uses the focal modulation mechanism to capture the long-distance dependence and contextual information within the image to enhance the model's capacity to identify defects that have subtle boundary characteristics; and 3) design the feature pyramid small object (SORP) for small object detection to improve

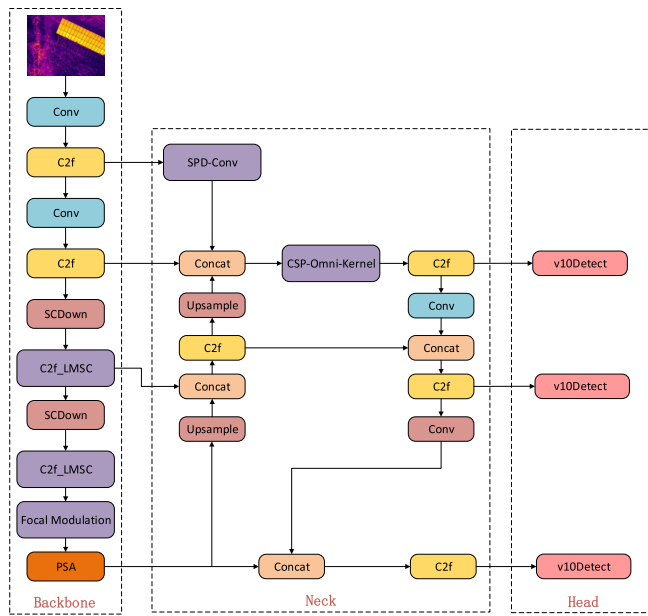


Fig. 6. LFS-YOLO.

the feature extraction capability of the model for small object defects. The enhanced model architecture is shown in Fig. 6.

A. Lightweight Multiscale Convolutional Module Design

The defects studied in this article have significant differences in size, shape, and so on. The C2f module in the pre-improved YOLOv10n only uses a single 3×3 convolutional kernel, which makes it difficult to adapt to the spatial feature extraction of each defect at the same time; moreover, the C2f module expands the channel dimensions by stacking repetitive bottleneck layers, which leads to the existence of channels in the extraction of defect features information redundancy. Therefore, it is necessary to design a module that can more efficiently fuse the information of defects of different sizes and reduce the redundant features to improve the detection performance and reliability of the model in practical applications.

Therefore, a multiscale lightweight convolution module C2f_LMSC is designed in this article. This module combines the advantage of Mobile-Net network in fusing multiple channel features using the group convolution concept and Ghost-Net network in reducing channel redundant features to improve the ability of the model to extract effective features of different scales from a large number of redundant feature maps and to reduce the number of parameters and model size. Multiple convolutional kernels of different scales are used in processing the input data, and the features are extracted by convolving the number of channels in each layer, half of which are directly waited for the subsequent fusion output without convolution operation and the other half are extracted by grouping operation (usually including 1×1 , 3×3 , 5×5 , and 7×7 convolution kernels, which are indicated by Φ in the figure), and the two parts of the feature maps are stacked for exchanging the channel information and feature fusion output by point-by-point convolution (1×1 Conv).

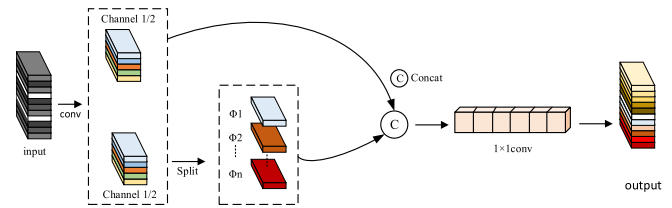


Fig. 7. C2f_LMSC network structure.

The defect information of different sizes is effectively fused, which improves the defect detection performance, at the same time reduces the number of parameters of the model and the size of the model, reduces the storage requirement of the model, and improves the reliability of the model in practical applications. The structure of the C2f_LMSC network is illustrated in Fig. 7.

B. Focused Modulation Network Design

The defects studied in this article have many boundary features that are not obvious, and the traditional SPPF module builds a pyramid structure by stacking the maximum pooling layers, although it can effectively extract multiscale features. When dealing with PV panel defects such as scenes with fuzzy boundaries and low-contrast features, its reliance on local spatial information aggregation is difficult to capture long-range semantic correlations across the region, and the fixed-window pooling operation is easy to lose important contextual cues, resulting in poor detection of such defects.

To improve the feature extraction ability of the model for defects with insignificant boundary features, this study adopts the focus modulation network Focal Nets to replace the original SPPF module. Compared with the SPPF module, which builds a spatial pyramid through a fixed-scale maximum pooling operation, Focal Nets implements a multibranch deep convolutional layer to encode different areas of visual contexts through the focus modulation mechanism, forming a contextual feature library with adjustable granularity. In particular, self-attention requires complex interactions of query keys and aggregation of query values for each query token with other tokens to compute attention scores and capture contexts, whereas focal modulation first aggregates spatial contexts into modulators with different granularities and then injects these modulators into query tokens in a query-dependent manner, which simplifies the interactions and aggregation operations and makes the network lighter. In addition, Focal Nets encode different ranges of visual contexts through deep convolutional layer stacking, selectively aggregate the context information into modulators of each query token through a gating mechanism, and then inject the aggregated modulators into each query token through affine transformation to improve the model's ability to capture key features of different defects and strengthen the feature response to the low-contrast defect regions. In particular, it is suitable for the difficult problem of detecting defects with inconspicuous boundary features on the surface of PV panels. The structure of the Focal Nets network is shown in Fig. 8.

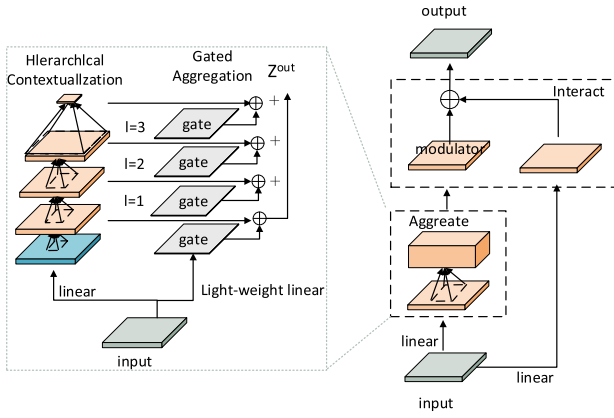


Fig. 8. Focal nets network structure.

C. Feature Pyramid Design

Aiming at the YOLOv10n model's leakage detection problem of small target defects, the SORP feature pyramid is designed in this study to improve the model's small target defect feature extraction capability. The original YOLOv10n's path aggregation feature pyramid network (PAFPN) constructs the P3–P5 detection layer through five downsampling operations, which weakens the shallow feature extraction ability, and multiple downsampling results in the loss of important feature detail information of small targets, making it ineffective on the original P3–P5 detection layer. A more common enhancement practice is to add a P2 detection layer to improve the detection ability of small targets, but this approach results in excessive model computation and more time-consuming postprocessing, which is not conducive to deployment in mobile devices.

In this study, the original PAFPN is enhanced by leveraging the P2 feature layer to access the P3 feature layer for fusion subsequent to spatial pyramid deformable (SPD)-Conv. SPD-Conv addresses the issue of detail loss since the process of downsampling in small target detection is converted from the spatial dimension to the depth dimension, thereby increasing the depth of the feature map. This approach retains as many defective features as possible without loss of information and avoids the information loss that occurs in traditional step-size convolution and pooling operations. The Omni-Kernel module is then enhanced by leveraging the cross-stage localized network [cross-stage partial network (CSP)] design concept. This approach involves the division of the feature map output from the convolutional layer into two segments. One segment enters the Omni-Kernel module to continue the convolution operation, while the other is directly fused with the feature map that has passed through the Omni-Kernel module to form the CSP-Omni-Kernel module. The Omni-Kernel module is designed to capture both the bottom-level detail information in the early stage and the high-level semantic information in the later stage. The output feature map can be efficiently downgraded, reducing the model's parameters and computational volume. This, in turn, enhances the performance of small target detection without generating excessive computational volume. The network structure of the CSP-Omni-Kernel module is shown in Fig. 9.

The Omni-Kernel module is comprised of three branches: the global branch, the large branch, and the local branch.

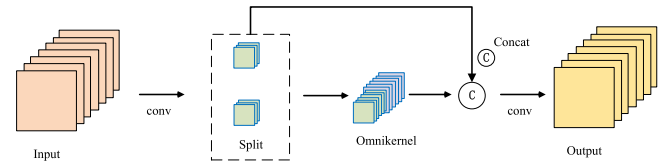


Fig. 9. CSP-Omni-Kernel network structure.

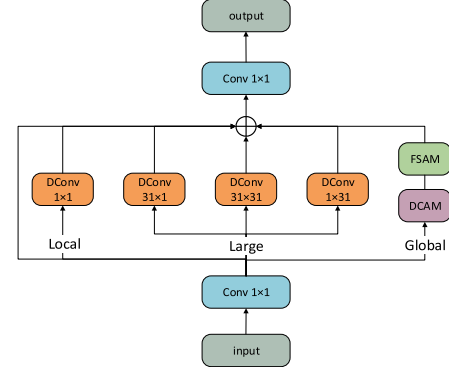


Fig. 10. Omni-Kernel network structure.

The results of these three branches are integrated through a process of addition, followed by convolution with a 1×1 kernel. This process captures that this fusion process integrates the global, large-scale local, and small-scale local features, enhancing the model's capacity to recognize defects as a whole and local details. This, in turn, can improve the model's adaptability to complex scenes. The structure of Omni-Kernel network is shown in Fig. 10.

V. RESULTS' TESTING AND ANALYSIS

A. Test Environment Configuration and Parameter Setting

The processing platform used in this experiment is a 10th generation Intel¹ Core² i9-10980XE CPU at 3.00 GHz with 64 GB of RAM and an NVIDIA Quadro RTX 5000 GPU with 16 GB of video memory. The software environment includes Cuda 12.0, Cudnn 8.7.0, conda 22.9.0, and Python 3.9.10, and the operating system is 64-bit Windows 10 Professional. During the model training process, the initial learning rate was set to 0.01, the batch size (Batch size) was 32, and the maximum number of iterations was set to 300. The weight parameters were optimized by stochastic gradient descent (SGD), and the decay strategy of the learning rate (lr) was described as follows:

$$lr = base_{lr} \cdot \left(1 - \frac{iter_{num}}{max_iterations}\right)^p \quad (1)$$

where $base_{lr}$ denotes the fundamental learning pace, max_iter indicates the highest iteration count, $iter$ represents the current iteration counter, and p signifies the polynomial decay parameter. For the purposes of this study, the initial learning rate was established at 0.001, the momentum value was configured to 0.9, the weight decay was configured to 1×10^{-4} , and the minimum threshold for learning rate updates was 0. Every

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TABLE II
MODEL IMPROVEMENT ABLATION TESTS

Test	C2f_LMSC	Focal Nets	SORP	mAP50%	FPS	Params/M	Model Size/MB
1	×	×	×	88.7	142.3	2.3	5.5
2	√	×	×	89.2	172.4	2.1	5.2
3	√	√	×	90.2	139.6	2.3	5.6
4	√	√	√	91.9	116.3	2.6	6.2

model underwent training according to these parameters. During the training process, the cross-entropy loss function was employed to evaluate the disparity between the algorithm's predicted probability distribution for the pixel category and the real probability distribution of the true label, which can be computed as follows:

$$\text{Loss} = \frac{1}{M} \sum_{i=1}^M \sum_{C=1}^N h(b_i) \log(p_{ic}) \quad (2)$$

where M represents the total number of pixels, N denotes the number of categories, i refers to the specific pixel being processed, C indicates the category under consideration, and b_i signifies the correct label for pixel i . The function h represents a probability density function that ranges from 0 to 1, equaling 1 when b_i matches c and 0 otherwise. Additionally, p_{ic} denotes the probability that pixel i belongs to category c , which is computed utilizing the sigmoid function applied to the forecast class score. Throughout the iterative training procedure, the value of the loss function was utilized to evaluate the model's training effectiveness. Subsequently, backpropagation was employed to optimize the weight parameters, aiming to minimize the model's loss value. Ultimately, the loss value was stabilized at a lower level, thereby finishing the model training.

B. Evaluation Indicators

To in an objective manner assess the effectiveness of the network model in classifying defects in infrared images of aerial PV and quantitatively analyze its performance, this study utilizes several metrics. These include the number of parameters, the number of giga floating-point operations per second (GFLOPs), inference speed [frames per second (FPS)], model size, precision, recall rate, average precision (AP), and mean AP (mAP). These metrics will be employed to evaluate the performance of the enhanced model. The definitions of these metrics are as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (3)$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (4)$$

$$\text{AP} = \int_0^1 P(R) dR \quad (5)$$

$$\text{mAP} = \frac{\sum_{i=1}^N \text{AP}_i}{N} \quad (6)$$

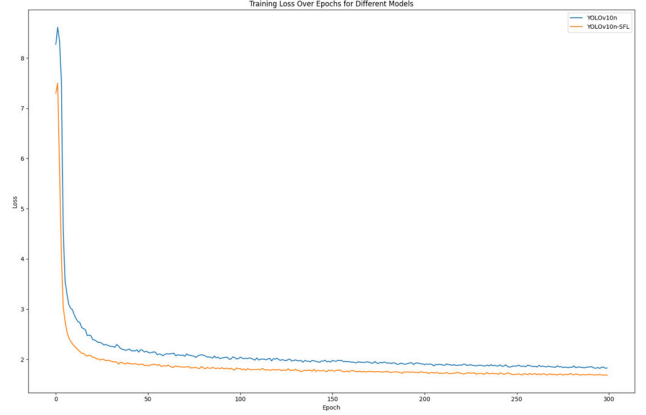


Fig. 11. Loss curve of model training process before and after improvement.

where N represents the total number of detection categories. The evaluation of accuracy and recall is conducted using a threshold of intersection over union (IoU) set at 0.5. The model assesses the samples across four categories: true positive (TP), true negative (TN), false positive (FP), and false negative (FN).

C. Ablation Test Results and Analysis

The curve of the loss function change during the training process of the improved model is shown in Fig. 11. By observing Fig. 11, it can be seen that in the initial stage of the model training, the convergence speed of the model is faster, which suggests that the model is capable of successfully capture the essential feature information from the data during the initial phase of training. With the increase in the number of training rounds, the loss function value of the model shows a gradual stabilization after about the 200th round, and the loss value converges and eventually stabilizes, indicating that the model gradually reaches the optimal state, and the weights with the smallest loss are verified as the optimal weights during the training process. The loss value of LFS-YOLO consistently remains lower than that of YOLOv10n throughout the entire training process, suggesting that the overall improvements have enhanced the model's performance.

In order to verify the effectiveness of each improvement on model performance enhancement, a series of ablation tests were designed for analysis. The environment and parameter settings were kept uniform for each test. The results of the ablation tests are shown in Table II.

As shown in Table II, LFS-YOLO enhances the model's detection accuracy by 3.2%, with a slight increase in computational load and the number of parameters. Test 1 uses

the unimproved YOLOv10n model with an average detection accuracy of 88.7%. Test 2 designs the multiscale lightweight convolutional module C2f_LMSC to replace part of the C2f module of the backbone network, which strengthens the ability of feature extraction for defects of different morphologies, also reduces the number of parameters of the model and the size of the model, and improves the speed of the model detection by 21% although the average accuracy of the detection is not much improved; Test 3 replaces the SPPF module of the original model with Focal Nets based on Test 2 and increases the speed of the model detection by 3.2% even though the computational volume and the size of the model have been reduced. For the SPPF module in the model, by replacing the self-attention module and using the focus modulation mechanism to capture the long-distance dependence and contextual information in the image, the average accuracy of the model detection is improved by 1%. Trial 4 joins the SORP for the detection of small targets, uses the P2 feature layer to get the information enriched with the small targets after the SPD-Conv, and improves the CSP idea to get the CSP-Omni-Kernel by the CSP-Omni-Kernel. The CSP-Omni-Kernel module is improved to obtain the CSP-Omni-Kernel module for feature integration, so as to efficiently learn the features from global to local and thus improve the average accuracy by 1.7% without generating too much computational effort. The improved LFS-YOLO model has a stronger ability to extract effective features for each defect, which can provide a reference for the subsequent defect detection of mobile terminals.

D. Comparative Analysis of Defect Key Feature Acquisition Capability Before and After Model Improvement

With the aim of visualizing the operational framework of the LFS-YOLO model proposed in this study for identifying common hot spot defects in PV panels, along with validating the efficacy of the model enhancement technique, the thermograms generated by the Grad-CAM visualization technique compare the regions in the images that contribute the most to the prediction results of the model before and after the improvement, which in turn helps to understand the model's decision-making rationale. Grad-CAM visualization results are shown in Fig. 12.

From the comparison in Fig. 12, it is obvious that the YOLOv10n model has some limitations in detecting diode failure and occlude, and its overall focus is not complete enough. It is easy to miss some key feature information when dealing with images where the target defects are too small, especially in the face of tiny solar cell defects, and the YOLOv10n model is often difficult to accurately capture these subtle defect features, resulting in poor detection results.

These problems are significantly improved in the LFS-YOLO model. The model extracts the tiny target feature information in the image by introducing SPD-Conv, which enables the model to dig deeper into the image details, thus accurately capturing the tiny defect features that are easily overlooked. In addition, the addition of the focus modulation mechanism enhances the ability of the model to capture

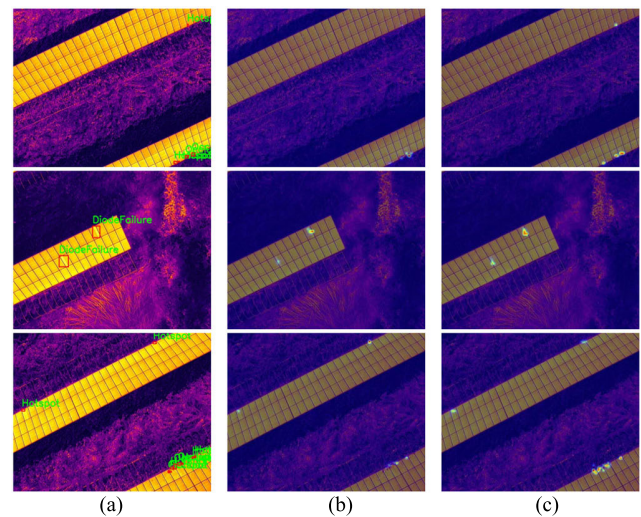


Fig. 12. Grad-CAM visualization before and after model improvement. (a) Lable. (b) YOLOv10n. (c) LFS-YOLO.

contextual information, and by capturing the long-range dependencies in the image, the model is able to capture the defect feature information more comprehensively, which in turn improves the focus on various types of defective regions and improves the situation that the original model could not focus on the defects of the tiny PV cells.

E. Comparative Analysis of Visualization of Defect Detection Effect Before and After Model Improvement

In order to verify the detection effect before and after the model improvement, three more representative infrared images of PV modules containing solar cell defects, occlude, and diode failure are selected for detection in the test set of this article's dataset, Fig. 13(a) shows the infrared image with the label of the real location of the defects, Fig. 13(b) shows the detection results of the YOLOv10n, and Fig. 13(c) shows the detection results of the LFS-YOLO. The detection results include defect range box, defect category, and defect confidence. From Fig. 13(1) and (2), it is obvious that the confidence of the model on the detection of all types of defects has been improved to different degrees, and from Fig. 13(3), it can be seen that the defect leakage caused by the small target problem has also been improved, which proves the effectiveness of the improvement.

VI. DISCUSSION

In recent years, with the advancement of deep learning, automatic defect detection of PV panels has solved the problems of low efficiency and high defect leakage rate of traditional manual inspection. However, there are still many challenges in this field. For example, existing studies are mostly based on convolutional operations with fixed sensory fields, which limits the ability to capture multiscale features under low computational constraints and does not consider the improvement of defect detection performance with unclear boundaries, in addition to the low accuracy of small target detection, which has been a key problem to be solved in this

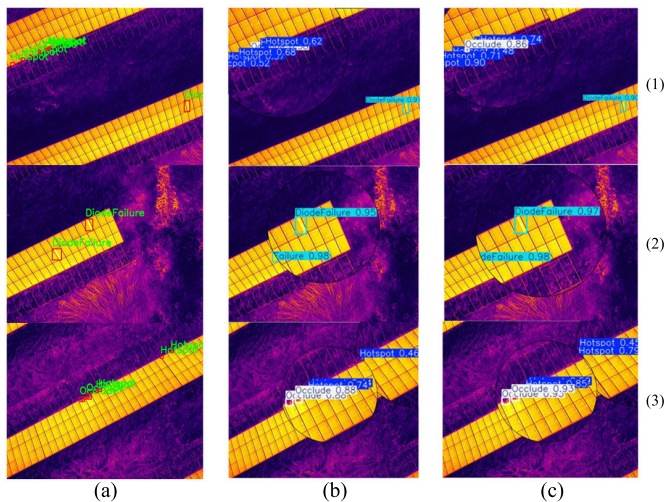


Fig. 13. Visualization of defect detection effect before and after model improvement. (a) Lable. (b) YOLOv10n. (c) LFS-YOLO. (1) Solar cell defects and diode failure. (2) Diode failure. (3) Occlude and solar cell defects.

field. To address these challenges, the LFS-YOLOv10 model proposed in this study shows certain advantages in the PV panel defect detection task. By introducing the C2f_LMSC module, the focal modulation network Focal Nets, and the SORP feature pyramid, the model achieves improved performance in multiscale feature extraction, boundary fuzzy defect identification, and small target detection. Experimental results show that the improved model achieves an average accuracy of 91.9% on the self-generated dataset, which is a 3.2% improvement over the baseline model.

Despite the above progress, this study still has some limitations. First, the current dataset is mainly collected from power plants under clear weather conditions, and factors such as rain, fog, and light variations can affect the quality of infrared images captured by drones, which may affect the robustness of the model. Second, although LFS-YOLO reduces the number of parameters and computational load, its computational requirements and inference speed may still be challenging for some embedded devices or resource-constrained environments. Finally, although the model introduces focal nets to improve the ability to detect defects with unclear boundaries, the model may still have difficulty in detecting defects that are extremely ambiguous or very small.

Therefore, future work will be carried out in the following three aspects: 1) constructing defect datasets in a multi-influence factor environment and introducing migration learning techniques to improve the model generalization ability; 2) using lightweight methods such as model pruning or knowledge distillation to reduce the model complexity while maintaining the detection accuracy; and 3) expanding the data of extremely fuzzy or tiny defects and exploring the optimization strategy to focus on the features of the defects that are difficult to identify and continuously improve the model performance.

VII. CONCLUSION

In this article, an LFS-YOLO model for detecting hot spot defects in PV is proposed, which solves the problem of poor

detection in drone aerial infrared images due to defects with different morphologies, inconspicuous boundary features, and the loss of features of small targets and at the same time ensures the model's ability to detect in real time. The main conclusions are summarized as follows.

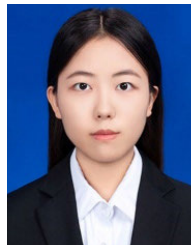
- 1) The C2f_LMSC module is introduced in the backbone network of the model. This module combines the ideas of lightweight Ghost-Net network, Mobile-Net network, and swarm convolution, aiming to extract key features of hot spot defects of PV panels more efficiently. With this integrated approach, the C2f_LMSC module is able to capture features at various scales and enhance the feature capability of the model.
- 2) For the purpose of improving the model's defect detection performance with inconspicuous boundary features, Focal Nets are used to replace the SPPF module in the original model. Focal Nets strengthen the model's ability to capture long-distance dependence and contextual information through the focus modulation mechanism, thus improving the recognition of defects with inconspicuous boundaries.
- 3) A feature pyramid specialized for small targets is designed to address the missed detection of small targets. By using the P2 feature layer and combining it with the SPD-Conv technique, the model is able to extract features that are rich in small target information. In addition, the CSP-Omni-Kernel module is designed, can effectively integrate the features, and can learn the information from global to local, which further improves the model's detection performance for small targets.

The experimental results show that the LFS-YOLO model achieves an average detection accuracy of 91.9% in the hot spot defect detection task of PV, which is a 3.2% improvement compared to the baseline model YOLOv10n. In addition, the LFS-YOLO model significantly improves the defect leakage detection phenomenon, showing good detection accuracy and generalization ability, which can provide technical support for PV panel defect detection based on drone thermal infrared sensors.

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